

- **Normal prompt:** “Tell me about renewable energy.”
The response to this prompt may include a lot of unnecessary information, or it may not focus on the aspect of renewable energy you’re interested in.
The key challenge here is that broad prompts lead to responses that may be too generic or unfocused. By adding specificity and structure, you can avoid this.
- **Engineered prompt:** “Explain the environmental benefits of renewable energy, with examples of how wind and solar power reduce carbon emissions.”

Prepare your input

To create an effective prompt, the first step is to prepare your input. This involves clearly defining what you want the AI to focus on and adding any necessary context to narrow down the response. Here’s an example.

- **Before prompt engineering:** “What is artificial intelligence?”
- **After engineering:** “Provide a brief explanation of artificial intelligence, focusing on its use in healthcare for diagnostic purposes.”

Notice how the engineered prompt includes the specific application of AI (healthcare) and the context (diagnostic purposes), which will guide the model towards a more targeted answer.

The important tips are:

- Be specific about the subject.
- Provide relevant background information.
- Define the focus area or scope of the response.

Plan the output

Planning the output involves thinking about what you expect in return from the AI, including the length, structure, or format of the response. An example is given below.

- **Without planning:** “Summarise the history of computers.”
- **With planning:** “Summarise the history of computers in three bullet points, focusing on key technological milestones from the 20th century.”

By planning the output, you can guide the AI towards generating a response that matches your needs—whether you want a list, a comparison, or a summary.

The key considerations here are:

- **Length:** Specify if you want a brief or detailed answer.
- **Format:** Ask for the output in bullet points, paragraphs, or a numbered list.
- **Tone/style:** Indicate if the response should be technical, formal, or conversational.

Play around with the output by testing different inputs

Experimenting with various prompts can help you understand how different phrasing or constraints affect the AI’s response. Slight changes in how you ask a question can lead to drastically different answers. Let’s look at two examples.

Example 1

- **First prompt:** “What are the benefits of exercise?”
- **Response:** A general answer listing a variety of benefits like weight loss, mental health, and heart health.
- **Tweaked prompt:** “What are the mental health benefits of regular exercise, specifically for people with anxiety?”
- **Response:** A more focused answer that centres on the connection between exercise and anxiety management.

Example 2

- **First prompt:** “Explain machine learning.”
- **Tweaked prompt:** “Explain machine learning in simple terms for a high school student.”

By changing the input, you can guide the AI towards different levels of complexity or different kinds of answers.

Here are a few useful tips for experimenting.

- Rephrase questions to see how different wording affects the output.
- Try adding constraints, such as asking for specific examples or a particular tone.
- Use multiple versions of a question to test the range of responses.

Efficient prompt engineering

Efficient prompt engineering saves time and helps you get the right answer without unnecessary back-and-forth. To do this, you can apply best practices that enhance clarity, specificity, and context. Here’s an example.

- **Inefficient prompt:** “How does the internet work?”
- **Efficient prompt:** “Explain how the internet works, focusing on the role of routers and IP addresses, and describe the process in 200 words.”

The efficient prompt is specific, sets clear expectations about the topic focus, and even defines the desired length of the response.

Here are the best practices.

- **Be explicit:** Clearly define the scope, length, and format of the response.
- **Iterate:** If the first response isn’t what you’re looking for, refine and test new prompts.
- **Test variations:** Explore different ways of asking the same question to find the most effective phrasing.